

Electronics & Instrumentation Engineering student with hands-on expertise in **Embedded Systems, Edge AI, IoT Hardware, Industrial Automation (PLC/HMI), and Software Development**. Seeking to apply skills in process automation, instrumentation, and intelligent control systems to build scalable, real-world solutions.

EDUCATION

Bannari Amman Institute of Technology
B.E. Electronics & Instrumentation Engineering
2022 – 2026 | Erode · CGPA: 7.67

Grace Higher Secondary School
12th (HSC)
2021 – 2022 · 81.33%

Govt. High School Alathur Thattapalayam
10th (SSLC)
2019 – 2020 · 74%

INSTRUMENTATION & AUTOMATION

Industrial Instrumentation

Process Control

PLC Programming

Ladder Logic

HMI

Pneumatic Systems

Sensor & Actuator Interfacing

Flow Calibration (YF-S201)

EMBEDDED & IOT

Embedded Systems Design

ESP32 / Raspberry Pi

GPIO / RFID / I2C / SPI

IoT Hardware Integration

Cloud Data Logging

e-Paper Display

OLED Integration

AI & SOFTWARE

Edge AI (1D CNN)

On-device Inference

C / Python / Java

HTML / CSS / JavaScript

Firmware Development

AREAS OF INTEREST

Industrial Instrumentation

Process Control & Automation

Embedded Systems

IoT Applications

Edge AI / ML

Software Development

PROJECTS

Smart Handwriting Learning Device

EMBEDDED SYSTEMS & EDGE AI DEVELOPER

National-Level Prototype Challenge – WAVES 2025 | 2nd Prize Winner

- Built a low-cost stencil device using conductive copper path tracing and real-time touch feedback for handwriting improvement.
- Implemented an offline **Edge AI model (1D CNN)** for stroke validation and pattern analysis on-device.
- Developed embedded control logic for timing, continuity detection, and real-time scoring.
- Integrated **ESP32 & Raspberry Pi** with e-paper display for offline, real-time interaction.
- GPIO-based continuity detection; fully offline, reliable system. Aligned with Digital India & NEP 2020.

ESP32

Raspberry Pi

1D CNN

Edge AI

e-Paper

Python

C

Automatic Milk Dispensing System

EMBEDDED & IOT DEVELOPER · FLOW CALIBRATION ENGINEER

Finalist – EW Challenge 2024

- Developed complete embedded firmware on microcontroller for automated liquid dispensing.
- Calibrated **YF-S201 flow sensor** for accurate liquid measurement.
- Integrated **RFID-based authentication** for secure user access.
- Enabled real-time **cloud-based data logging**, transaction tracking, and system diagnostics.

Microcontroller

RFID

YF-S201

IoT

Cloud Logging

C

Intelligent Pneumatic Control System

DESIGN & AUTOMATION ENGINEER

JASC 2024 – CAD Design & Industrial Automation Challenge

- Designed and programmed a **PLC-based pneumatic automation** setup for industrial applications.
- Prepared CAD models using **Fusion 360** and tuned system parameters for smooth actuation.
- Improved operational efficiency, energy efficiency, reliability, and system safety.

PLC

Ladder Logic

Pneumatics

Siemens TIA Portal

Fusion 360

Automated Environmental Monitoring System

CIRCUIT & SYSTEM DEVELOPER

Multi-Sensor IoT System

- Designed a multi-sensor system using **ESP32** with gas, temperature, and humidity sensors.
- Automated control of fans and relays to maintain stable plant conditions.
- Developed firmware for real-time display on **OLED** and cloud data logging.

ESP32

Gas Sensor

DHT Sensor

IoT

OLED

Relay Control

TOOLS & PLATFORMS

- > Siemens TIA Portal
- > Arduino IDE
- > Proteus
- > Fusion 360
- > LabVIEW
- > KiCad

SOFT SKILLS

- Problem Solving
- Leadership
- Teamwork
- Critical Thinking
- Time Management

LANGUAGES

- > English (Read, Write, Speak)
- > Tamil (Read, Write, Speak)

AI-Powered Smart FOB & Road Safety System

IDEATION · SYSTEM ARCHITECTURE · PRESENTATION

- 🏆 National Road Safety Hackathon 2025 – NHAI & HOAI | **National Semi-Finalist**
- Proposed an **AI-powered Smart FOB** and road safety system using IoT, motion sensors, and smart signage.
- Designed a scalable real-time alert architecture to reduce jaywalking and enhance pedestrian safety.
- Selected for National Semi-Final by NHAI & HOAI.

- AI
- IoT
- Motion Sensors
- Smart Signage
- System Architecture

Lab Equipment Booking & Maintenance Tracker

FRONTEND / SOFTWARE DEVELOPER

- 💻 Full Web Application (Solo)
- Designed a responsive, role-based web app enabling students to book lab equipment and raise maintenance tickets.
- Admins can manage bookings, resolve issues, and monitor usage history.
- Implemented tabbed dashboards, dynamic form handling, simulated authentication, and notifications using pure client-side technologies.

- HTML
- CSS
- JavaScript
- Responsive Design
- Role-Based Auth

ACHIEVEMENTS

- 🏆 **2nd Prize – Innovate2Educate Challenge, WAVES Summit 2025**
Showcased Smart Handwriting Learning Device supporting Digital India goals.

- 🏁 **National Semi-Finalist – Road Safety Hackathon 2025 (NHAI & HOAI)**
Selected for proposing an AI-powered Smart FOB and pedestrian safety system using IoT and real-time alerts.

- 🏆 **Finalist – EW Challenge 2024**
Impressed evaluators with precise flow measurement, RFID authentication, and IoT-based automation in milk dispensing system.

I hereby declare that the above-mentioned details are true to the best of my knowledge.
Abhivadan J